

# Continuum Mechanics HW4

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*Technion*

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# 1 Flamant Solution for a Point Tangential Stress

## 1.1 radial only

The stress  $Q$  is applied purely in the radial direction so none of it is applied in the angular direction, so  $\sigma_{\theta r} = \sigma_{\theta\theta} = 0$ .

## 1.2 Airy stress

Since the solutions we need are equilibrium solutions, we have  $\nabla \cdot \sigma = 0$ . Therefore integrating over half a sphere like in the TA session

$$0 = \int_A dA \nabla \cdot \sigma = \int_0^R dr \int_{-\pi/2}^{\pi/2} d\theta r \nabla \cdot \sigma \quad (1)$$

$$= \int_{-\pi/2}^{\pi/2} d\theta R \hat{\mathbf{r}} \cdot \sigma + \int_{-R}^R (-\hat{\mathbf{x}}) \cdot \sigma dy \quad (2)$$

$$= \int_{-\pi/2}^{\pi/2} d\theta R \sigma_{rr} \hat{\mathbf{r}} - \int_{-R}^R \sigma_{xy} \hat{\mathbf{y}} dy \quad (3)$$

$$= \int_{-\pi/2}^{\pi/2} d\theta R \sigma_{rr} \hat{\mathbf{r}} + Q \int_{-R}^R \delta(r) \hat{\mathbf{y}} dy \quad (4)$$

so

$$Q \int_{-R}^R \delta(r) \hat{\mathbf{y}} dy = - \int_{-\pi/2}^{\pi/2} d\theta R \sigma_{rr} \hat{\mathbf{r}} \quad (5)$$

$$= -R \left( \int_{-\pi/2}^{\pi/2} d\theta \sigma_{rr} \cos \theta \hat{\mathbf{x}} + \int_{-\pi/2}^{\pi/2} d\theta \sigma_{rr} \sin \theta \hat{\mathbf{y}} \right) \quad (6)$$

which we can now split into two equations by unit vector

$$Q = -R \int_{-\pi/2}^{\pi/2} d\theta \sigma_{rr} \sin \theta \quad (7)$$

$$0 = R \int_{-\pi/2}^{\pi/2} \sigma_{rr} \cos \theta \quad (8)$$

Since  $Q$  is a constant the integral must resolve to a quantity that includes  $1/r$  in it somewhere, meaning

$$\sigma_{rr} \propto \frac{F(\theta)}{r} \quad (9)$$

where  $F$  is some function. Using the expression from the training

$$\sigma_{rr} = \frac{1}{r} \frac{\partial \chi}{\partial r} + \frac{1}{r^2} \frac{\partial^2 \chi}{\partial \theta^2} \propto \frac{1}{r} \implies F(\theta) = \frac{\partial \chi}{\partial r} + \frac{1}{r} \frac{\partial^2 \chi}{\partial \theta^2} \quad (10)$$

and continuing with the training we find that we must have

$$\chi = (A \cos \theta + B \sin \theta) r \theta \quad (11)$$

and now derivatives

$$\sigma_{rr} = \frac{1}{r} \frac{\partial \chi}{\partial r} + \frac{1}{r^2} \frac{\partial^2 \chi}{\partial \theta^2} \quad (12)$$

$$= \dots = \frac{-2A \sin \theta + 2B \cos \theta}{r} \quad (13)$$

again from the training. Throwing this into the expressions we found

$$0 = R \int_{-\pi/2}^{\pi/2} \frac{-2A \sin \theta + 2B \cos \theta}{R} \cos \theta \quad (14)$$

$$= \int_{-\pi/2}^{\pi/2} (-2A \sin \theta + 2B \cos \theta) \cos \theta \quad (15)$$

$$= \int_{-\pi/2}^{\pi/2} 2B \cos^2 \theta \implies B = 0 \quad (16)$$

and

$$Q = 2A \int_{-\pi/2}^{\pi/2} d\theta \sin^2 \theta \quad (17)$$

$$= 2A \cdot \pi/2 = \pi A \implies A = \frac{Q}{\pi} \quad (18)$$

so in total

$$\chi = \frac{Q}{\pi} \cos \theta r \theta, \quad \sigma_{rr} = -\frac{2Q}{\pi} \frac{\sin \theta}{r} \quad (19)$$

### 1.3 equal-stress contours

for constant stress

$$\frac{2Q}{\pi} \frac{\sin \theta}{r} \quad (20)$$

so as in the training we define

$$D = \frac{Q}{\pi C} \quad (21)$$

and

$$\frac{1}{D} = \frac{2 \sin \theta}{r^2} r^2 = 2Dr \sin \theta \quad (22)$$

converting to cartesian for humans

$$x^2 + y^2 - 2Dy = 0 \implies x^2 + y^2 - 2Dy + D^2 = D^2 \implies x^2 + (y - D)^2 = D^2 \quad (23)$$

giving a circle around  $y = D$ . For the other circle, we notice that setting  $D \rightarrow -D$  changes nothing, so  $y = -D$  is also a circle center.

### 1.4 general tangential stress

in cartesian from radial

$$\sigma_{xx} = \sigma_{rr} \cos^2 \theta = \frac{2Q}{\pi} \frac{x^2 y}{(x^2 + y^2)^2} \quad (24)$$

$$\sigma_{yy} = \sigma_{rr} \sin^2 \theta = \frac{2Q}{\pi} \frac{y^3}{(x^2 + y^2)^2} \quad (25)$$

$$\sigma_{xy} = \sigma_{rr} \sin \theta \cos \theta = \frac{2Q}{\pi} \frac{xy^2}{(x^2 + y^2)^2} \quad (26)$$

but since we want a general stress distribution, instead of  $Q$  we have  $q(s)$  so

$$\sigma_{xx} = \frac{2x^2}{\pi} \int_{-\infty}^{\infty} \frac{(y-s)q(s)}{(x^2 + (y-s)^2)^2} \quad (27)$$

$$\sigma_{yy} = \frac{2}{\pi} \int_{-\infty}^{\infty} \frac{(y-s)^3 q(s)}{(x^2 + (y-s)^2)^2} \quad (28)$$

$$\sigma_{xy} = \frac{2x}{\pi} \int_{-\infty}^{\infty} \frac{(y-s)^2 q(s)}{(x^2 + (y-s)^2)^2} \quad (29)$$